



FATE MONTHLY REPORT

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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
Annex
October 2023



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wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	693	215	3
	4.2 < ws < 7.6	411	597	224
	ws > 7.6	50	274	1618
Sample size: 6588; PC=71.2%; EBD=1.3%; POD1=60.1%; POD2=55.0%; POD3=87.7%				

Table 1: Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}		Observations		
		ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	1087	82	0
	4.2 < ws < 7.6	67	944	29
	ws > 7.6	0	60	1816
Sample size: 6588; PC=94.2%; EBD=0.0%; POD1=94.2%; POD2=86.9%; POD3=98.4%				

Table 2: Contingency table for wind speed (ms^{-1}) processed with AR (1H)

0.1 Title for Contingency tables of Night 1

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wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
ws < 12.0	1561	26	0
Model data 12.0 < ws < 18.0	17	373	8
ws > 18.0	0	1	91
Sample size: 3348; PC=97.5%; EBD=0.0%; POD1=98.9%; POD2=93.2%; POD3=91.9%			

Table 3: Contingency table for wind speed (ms^{-1}) processed with AR (1H)

FATE - Monthly Report

wind direction degree		North	East	South	Observations	West
Model data	North	2318	101	14		62
	East	49	133	38		0
	South	5	36	374		32
	West	99	17	1		75

Sample size: 6588; PC=86.5%; EBD=1.1%; POD1=93.8%; POD2=46.3%; POD3=87.6%; POD4=44.4%

Table 4: Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	2452	30	0		5
	East	14	243	5		0
	South	0	14	418		0
	West	5	0	4		164

Sample size: 6588; PC=97.7%; EBD=0.0%; POD1=99.2%; POD2=84.7%; POD3=97.9%; POD4=97.0%

Table 5: Contingency table for wind direction (degree) in standard configuration

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relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	581	350
	7.1 < rh < 15.3	538	1359
	rh > 15.3	50	261
Sample size: 6588; PC=65.7%; EBD=1.5%; POD1=49.7%; POD2=69.0%; POD3=80.6%			

Table 6: Contingency table for relative humidity (percent) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	1083	74
	7.1 < rh < 15.3	85	1857
	rh > 15.3	1	39
Sample size: 6588; PC=93.7%; EBD=0.1%; POD1=92.6%; POD2=94.3%; POD3=93.9%			

Table 7: Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent	Observations		
	rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	1997	0
	50.0 < rh < 70.0	1	0
	rh > 70.0	0	0
Sample size: 3348; PC=99.9%; EBD=0.0%; POD1=99.9%; POD2=NaN%; POD3=NaN%			

Table 8: Contingency table for relative humidity (percent) processed with AR (1H)

FATE - Monthly Report

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1168	230	0
	1.9 < pwv < 3.5	51	1143	325
	pwv > 3.5	0	76	867
Sample size: 6588; PC=82.3%; EBD=0.0%; POD1=95.8%; POD2=78.9%; POD3=72.7%				

Table 9: Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	1198	30	0
	1.9 < pwv < 3.5	21	1410	22
	pwv > 3.5	0	9	1170
Sample size: 6588; PC=97.9%; EBD=0.0%; POD1=98.3%; POD2=97.3%; POD3=98.2%				

Table 10: Contingency table for precipitable water vapor (mm) processed with AR (1H)

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total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	452	244	102
	0.6 < see < 0.8	162	290	296
	see > 0.8	43	123	258
Sample size: 5184; PC=50.8%; EBD=7.4%; POD1=68.8%; POD2=44.1%; POD3=39.3%				

Table 11: Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.0

total seeing arcsec		Observations		
		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	422	105	0
	0.6 < see < 0.8	50	498	53
	see > 0.8	0	52	1100
Sample size: 5184; PC=88.6%; EBD=0.0%; POD1=89.4%; POD2=76.0%; POD3=95.4%				

Table 12: Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.0

FATE - Monthly Report

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	610	0	116
	0.6 < see < 0.8	4	636	116
	see > 0.8	43	21	540
Sample size: 5184; PC=90.7%; EBD=8.1%; POD1=92.8%; POD2=96.8%; POD3=69.9%				

Table 13: Contingency table for total seeing (arcsec) in standard configuration and accuracy 0.24

total seeing		Observations		
arcsec		see < 0.6	0.6 < see < 0.8	see > 0.8
Model data	see < 0.6	472	0	0
	0.6 < see < 0.8	0	653	0
	see > 0.8	0	2	1153
Sample size: 5184; PC=99.9%; EBD=0.0%; POD1=100.0%; POD2=99.7%; POD3=100.0%				

Table 14: Contingency table for total seeing (arcsec) processed with AR (1H) and accuracy 0.24

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coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	760	164	0
	$3.4 < \tau < 5.8$	249	405	106
	$\tau > 5.8$	28	113	197
Sample size: 5184; PC=67.4%; EBD=1.4%; POD1=73.3%; POD2=59.4%; POD3=65.0%				

Table 15: Contingency table for coherence time (ms) in standard configuration and accuracy 0.0

coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1003	53	0
	$3.4 < \tau < 5.8$	34	592	29
	$\tau > 5.8$	0	37	274
Sample size: 5184; PC=92.4%; EBD=0.0%; POD1=96.7%; POD2=86.8%; POD3=90.4%				

Table 16: Contingency table for coherence time (ms) processed with AR (1H) and accuracy 0.0

coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	975	12	66
	$3.4 < \tau < 5.8$	34	605	66
	$\tau > 5.8$	28	65	237
Sample size: 5184; PC=89.9%; EBD=4.6%; POD1=94.0%; POD2=88.7%; POD3=64.2%				

Table 17: Contingency table for coherence time (ms) in standard configuration and accuracy 1.22

coherence time		Observations		
ms		$\tau < 3.4$	$3.4 < \tau < 5.8$	$\tau > 5.8$
Model data	$\tau < 3.4$	1037	0	2
	$3.4 < \tau < 5.8$	0	681	2
	$\tau > 5.8$	0	1	301
Sample size: 5184; PC=99.9%; EBD=0.1%; POD1=100.0%; POD2=99.9%; POD3=98.7%				

Table 18: Contingency table for coherence time (ms) processed with AR (1H) and accuracy 1.22

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	540	316	171
	0.6 < glf < 0.7	247	357	239
	glf > 0.7	101	62	79
Sample size: 5184; PC=46.2%; EBD=12.9%; POD1=60.8%; POD2=48.6%; POD3=16.2%				

Table 19: Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	808	117	1
	0.6 < glf < 0.7	79	530	82
	glf > 0.7	1	88	406
Sample size: 5184; PC=82.6%; EBD=0.1%; POD1=91.0%; POD2=72.1%; POD3=83.0%				

Table 20: Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.0

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	779	76	188
	0.6 < glf < 0.7	8	648	134
	glf > 0.7	101	11	301
Sample size: 5184; PC=81.8%; EBD=13.7%; POD1=87.7%; POD2=88.2%; POD3=48.3%				

Table 21: Contingency table for ground layer fraction (dimensionless) in standard configuration and accuracy 0.14

ground layer fraction dimensionless		Observations		
		glf < 0.6	0.6 < glf < 0.7	glf > 0.7
Model data	glf < 0.6	887	6	2
	0.6 < glf < 0.7	0	728	2
	glf > 0.7	1	1	487
Sample size: 5184; PC=99.5%; EBD=0.1%; POD1=99.9%; POD2=99.0%; POD3=99.2%				

Table 22: Contingency table for ground layer fraction (dimensionless) processed with AR (1H) and accuracy 0.14

wind speed ms^{-1}	Observations		
	ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	221	67
	4.2 < ws < 7.6	424	979
	ws > 7.6	34	236
Sample size: 6588; PC=72.1%; EBD=0.9%; POD1=32.5%; POD2=76.4%; POD3=82.5%			

Table 23: Contingency table for wind speed (ms^{-1}) in standard configuration

wind speed ms^{-1}	Observations		
	ws < 4.2	4.2 < ws < 7.6	ws > 7.6
Model data	ws < 4.2	593	90
	4.2 < ws < 7.6	86	1129
	ws > 7.6	0	63
Sample size: 6588; PC=91.8%; EBD=0.0%; POD1=87.3%; POD2=88.1%; POD3=95.6%			

Table 24: Contingency table for wind speed (ms^{-1}) processed with AR (1H)

0.2 Title for Contingency tables of Day 1

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wind speed ms^{-1}	Observations		
	ws < 12.0	12.0 < ws < 18.0	ws > 18.0
ws < 12.0	1370	38	0
Model data 12.0 < ws < 18.0	26	441	14
ws > 18.0	0	9	179
Sample size: 3348; PC=95.8%; EBD=0.0%; POD1=98.1%; POD2=90.4%; POD3=92.7%			

Table 25: Contingency table for wind speed (ms^{-1}) processed with AR (1H)

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wind direction degree		North	East	South	Observations	West
Model data	North	1922	96	0		57
	East	23	77	8		0
	South	0	0	254		83
	West	582	0	10		584

Sample size: 6588; PC=76.8%; EBD=0.0%; POD1=76.1%; POD2=44.5%; POD3=93.4%; POD4=80.7%

Table 26: Contingency table for wind direction (degree) in standard configuration

wind direction degree		North	East	South	Observations	West
Model data	North	2438	25	0		53
	East	15	148	0		0
	South	0	0	265		8
	West	74	0	7		663

Sample size: 6588; PC=95.1%; EBD=0.0%; POD1=96.5%; POD2=85.5%; POD3=97.4%; POD4=91.6%

Table 27: Contingency table for wind direction (degree) in standard configuration

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relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	326	194
	7.1 < rh < 15.3	318	1749
	rh > 15.3	45	337
Sample size: 6588; PC=72.9%; EBD=1.1%; POD1=47.3%; POD2=76.7%; POD3=81.8%			

Table 28: Contingency table for relative humidity (percent) in standard configuration

relative humidity percent	Observations		
	rh < 7.1	7.1 < rh < 15.3	rh > 15.3
Model data	rh < 7.1	617	63
	7.1 < rh < 15.3	72	2147
	rh > 15.3	0	70
Sample size: 6588; PC=93.8%; EBD=0.0%; POD1=89.6%; POD2=94.2%; POD3=95.9%			

Table 29: Contingency table for relative humidity (percent) processed with AR (1H)

relative humidity percent	Observations		
	rh < 50.0	50.0 < rh < 70.0	rh > 70.0
Model data	rh < 50.0	2055	0
	50.0 < rh < 70.0	0	0
	rh > 70.0	0	0
Sample size: 3348; PC=100.0%; EBD=0.0%; POD1=100.0%; POD2=NaN%; POD3=NaN%			

Table 30: Contingency table for relative humidity (percent) processed with AR (1H)

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	747	184	3
	1.9 < pwv < 3.5	185	1256	355
	pwv > 3.5	0	125	1002
Sample size: 6588; PC=77.9%; EBD=0.1%; POD1=80.2%; POD2=80.3%; POD3=73.7%				

Table 31: Contingency table for precipitable water vapor (mm) in standard configuration

precipitable water vapor mm		Observations		
		pwv < 1.9	1.9 < pwv < 3.5	pwv > 3.5
Model data	pwv < 1.9	898	22	0
	1.9 < pwv < 3.5	34	1530	31
	pwv > 3.5	0	13	1329
Sample size: 6588; PC=97.4%; EBD=0.0%; POD1=96.4%; POD2=97.8%; POD3=97.7%				

Table 32: Contingency table for precipitable water vapor (mm) processed with AR (1H)